

CASE 30

CONGENITAL ASPLENIA

The role of the spleen in immunity.

The adaptive immune response occurs mainly in the peripheral lymphoid tissues—the lymph nodes, the gut-associated lymphoid tissue, and the spleen (Fig. 30.1). Pathogens and their secreted antigens are trapped in these tissues and presented to the naive lymphocytes that constantly pass through. Microorganisms that enter the body through the skin or the lungs drain to regional lymph nodes, where they

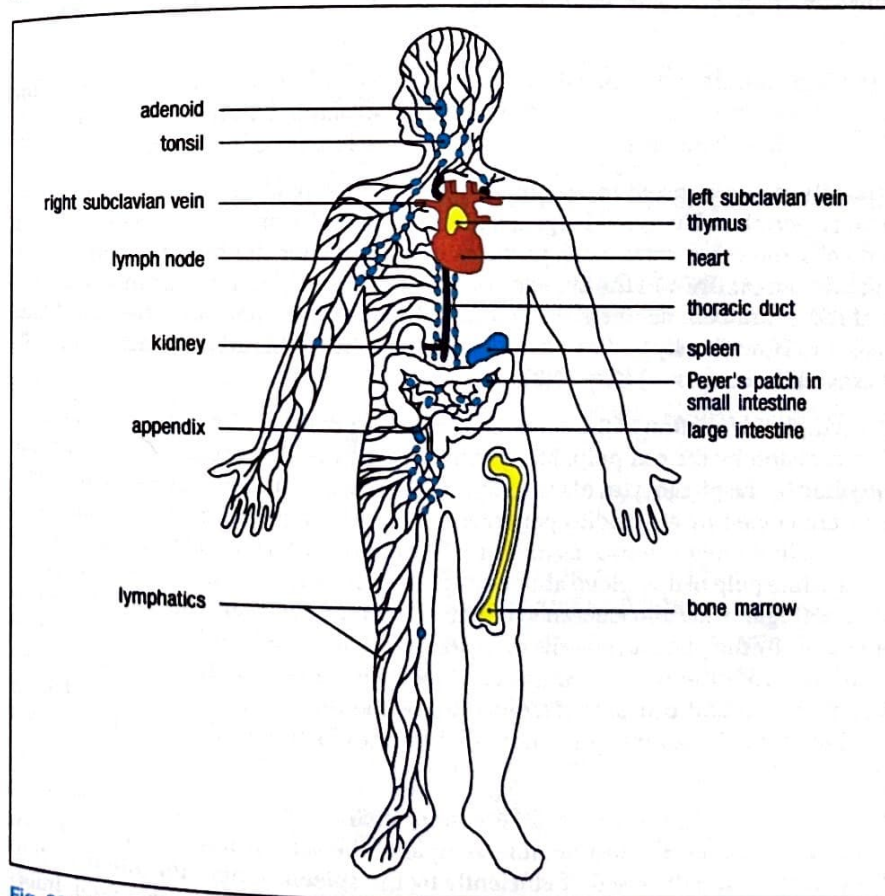


Fig. 30.1 The distribution of lymphoid tissues in the body. Lymphocytes arise from stem cells in bone marrow, and differentiate in the central lymphoid organs (yellow)—B cells in bone marrow and T cells in the thymus. They migrate from these tissues through the bloodstream to the peripheral lymphoid tissues (blue)—the lymph nodes, spleen, and mucosa-associated lymphoid tissues such as tonsils, Peyer's patches, and appendix. These are the sites of lymphocyte activation by antigen. Lymphatics drain extracellular fluid as lymph through the lymph nodes and into the thoracic duct, which returns the lymph to the bloodstream by emptying into the left subclavian vein. Lymphocytes that circulate in the bloodstream enter the peripheral lymphoid organs, and are eventually carried by lymph to the thoracic duct, where they reenter the bloodstream.

TOPICS BEARING ON THIS CASE:

Circulation of lymphocytes through peripheral lymphoid tissues

Toxoid vaccines

Role of the marginal zone of the spleen

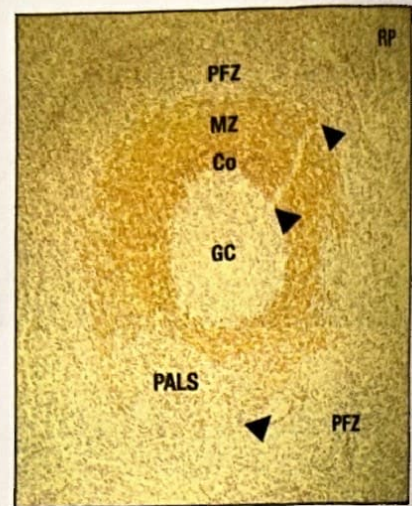
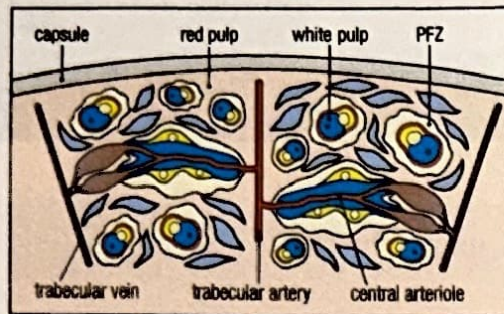
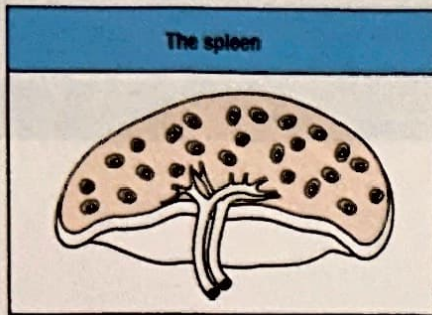


Fig. 30.2 Schematic views and light micrograph of a section of human spleen. The schematic at left shows that the spleen consists of red pulp (pink areas), which is a site of red blood cell destruction, interspersed with the lymphoid white pulp. An enlargement of an area of white pulp is shown in the middle panel. The follicle (yellow) and T-cell areas (blue) are surrounded by the

perifollicular zone (PFZ) (pale yellow). The light micrograph on the right shows a transverse section of white pulp immunostained for mature B cells. Co, follicular B-cell corona; GC, germinal center (activated B cells); MZ, marginal zone (B cells); PALS, periarteriolar lymphoid sheath; RP, red pulp; arrowheads, central arteriole. Photograph courtesy of N.M. Milicevic.

stimulate an immune response. Microorganisms and food antigens that enter the gastrointestinal tract are collected in the gut-associated lymphoid tissue. Microbes that enter the bloodstream stimulate an immune response in the spleen.

The spleen is organized to accomplish two functions (Fig. 30.2). In addition to being a peripheral lymphoid organ, it filters the blood, removing aged or abnormal red cells and other extraneous particles that may enter the bloodstream, including microorganisms. In the absence of a functioning spleen, the aged and abnormal red blood cells can be seen in a peripheral blood smear as pitted red blood cells and Howell-Jolly bodies (nuclear remnants in red blood cells that are usually removed by the spleen) (Fig. 30.3).

The lymphoid function of the spleen is performed in the white pulp, and the filtration function by the red pulp. Many microorganisms are recognized directly and engulfed by the phagocytes of the red pulp. Others are not removed efficiently until they are coated by antibodies generated in the white pulp. In experimental animals, an immune response (as measured by antibody formation) can be detected in the white pulp of the spleen about 4 days after the intravenous injection of a dose of microorganisms. The clearance of antibody- and complement-coated bacteria or viruses by the phagocytic cells of the red pulp of the spleen is very rapid. Rapid clearance from the blood is important because it prevents these bacteria from disseminating and causing infections of the meninges (meningitis), the kidney (pyelonephritis), the lung (pneumonia), the bones (osteomyelitis), or other distant anatomical sites.

Bacteria enter the bloodstream all the time, such as when we brush our teeth or when we have a local infection, for example of the skin or middle ear. Normally, these bacteria are disposed of efficiently by the spleen. When, for one reason or another, the spleen is not present or is non-functional, serious, even fatal, infections occur.

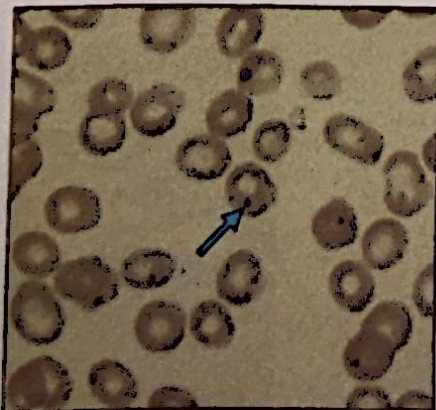


Fig. 30.3 Red blood cells with Howell-Jolly bodies. The arrow marks a Howell-Jolly body in a red blood cell. These inclusions are formed by the retention of nuclear remnants in red cells, which are usually removed by the spleen but can be found in patients with asplenia or significantly decreased splenic function. (From (<http://www.ezhemeonc.com/index.php/hematological-disorders/>) with permission from J. Hamdan, University of Illinois at Chicago College of Medicine.)

The case of Susan Vanderveer: a fatality because of an absent spleen.

Mr and Mrs Vanderveer owned a farm in the Hudson Valley in lower New York State. They were both descended from Dutch settlers who came to the Hudson Valley in the mid-17th century. There were multiple consanguineous marriages among their ancestors, and Mr and Mrs Vanderveer were distantly related to each other. At the time of this case, they had five children—three girls and two boys. Their youngest daughter, Susan, was 10 months old when she developed a cold, which lasted for 2 weeks. On the 14th day of her upper respiratory infection, she became sleepy and felt very warm. Her mother found that her temperature was 41.7°C. When Susan developed convulsive movements of her extremities, she was rushed to the emergency room but she died on the way to the hospital. Post-mortem cultures of blood were obtained, and also from her throat and cerebrospinal fluid. All the cultures grew *Haemophilus influenzae*, type b. At autopsy Susan was found to have no spleen.

At the time of Susan's death her 3-year-old sister, Betsy, also had a fever, of 38.9°C. She complained of an earache, and her eardrums were found to be red. She had no other complaints and no other abnormalities were detected on physical examination. Her white blood count was 28,500 cells μL^{-1} (very elevated). Cultures from her nose, throat, and blood grew out *H. influenzae*, type b. She was given ampicillin intravenously for 10 days in the hospital and was then sent home in good health. Her cultures were negative at the time of discharge from the hospital. She was seen by a pediatrician on three occasions during the following year for otitis media (inflammation of the middle ear), pneumonia, and mastoiditis (inflammation of the mastoid bone behind the ear).

David, Susan's 5-year-old brother, had been admitted to the hospital at 21 months of age with meningitis caused by *Streptococcus pneumoniae*. He had responded well to antibiotic therapy and had been discharged. Another occurrence of pneumococcal meningitis at 27 months of age had also been followed by an uneventful recovery after antibiotics. He had had pneumonia at age 3½ years. At the time of Susan's death he was well.

The two other children of the Vanderveers, a girl aged 8 years and a newborn male, were in good health.

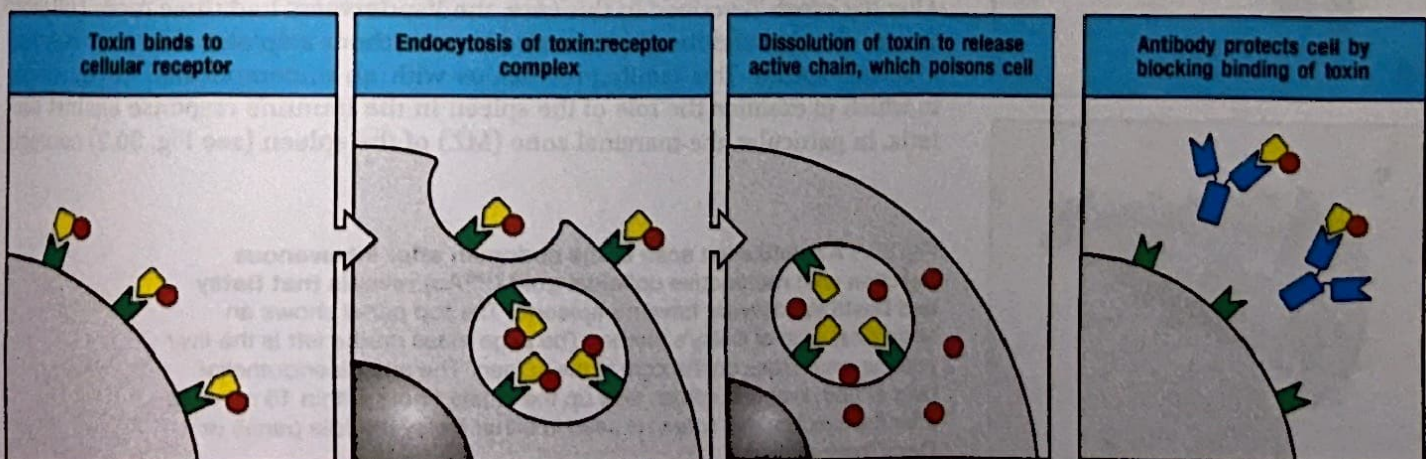
All the Vanderveer children had received routine immunization at ages 2, 4, and 6 months with tetanus and diphtheria toxoids and killed *Bordetella pertussis* to protect against tetanus, diphtheria, and whooping cough, which are three potentially fatal diseases caused by bacterial toxins (Fig. 30.4). All of the Vanderveer children had protective antibody responses against these antigens, indicating a normal

Susan Vanderveer, age
10 months, dead on arrival in
Emergency.

Betsy Vanderveer, age 3 years,
presents with severe
H. influenzae infection.

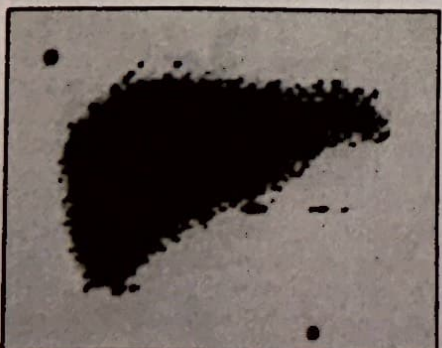
Fig. 30.4 Neutralization by antibodies protects cells from toxin action.

Secreted bacterial toxins usually contain several distinct moieties. One piece of the toxin must bind a cellular receptor, which allows the molecule to be internalized. A second part of the toxin molecule then enters the cytoplasm and poisons the cell. In some cases, a single molecule of toxin can kill a cell. Antibodies that inhibit toxin binding can prevent, or neutralize, these effects. Protective antibodies can be generated by subcutaneous immunization with toxoids. Toxoids are toxins rendered harmless by treatment with denaturing agents, such as formalin, which destroy their toxicity but not their ability to generate neutralizing antibodies. In the case of the DPT vaccine, the killed *Bordetella pertussis* cells act as an adjuvant, which enhances the immune response to all components of the vaccine by delivering activating signals to antigen-presenting cells.



Normal antibody response
to diphtheria, tetanus, and
pertussis.

Defective antibody response to
polysaccharide pneumococcal
vaccine.



immune response to T-dependent antigens. However, when they were immunized with polysaccharide vaccine against *S. pneumoniae* that contains capsular antigens from 23 distinct strains of *S. pneumoniae*, neither David nor Betsy were able to mount a good antibody response, whereas their two healthy siblings did. This indicated that Betsy and David have a defect in mounting antibody responses against polysaccharide antigens.

All the children and their parents were injected intravenously with radioactive colloidal gold (^{198}Au), which is taken up by the reticuloendothelial cells of the liver and spleen within 15 minutes after the injection. A scintillation counter then scans the abdomen for radioactive gold. The pattern of scintillation revealed that Betsy and David had no spleens (Fig. 30.5).

Asplenia and splenectomy.

Significant impairment of splenic function can be either congenital, where the spleen is absent or dysfunctional at birth, or acquired as a result of conditions that damage the spleen such as trauma or sickle cell anemia, and that often lead to its surgical removal (splenectomy). Congenital asplenia is further divided into two main categories. The more prevalent is syndromic asplenia, in which the lack of splenic tissue is part of a more complex genetic syndrome affecting other systems as well. In these syndromes splenic defects are usually associated with significant heart defects and heterotaxia, a condition in which malformations arise as a result of lateralization defects of organs in the thorax and the abdomen. Several human genes that have important roles in directing lateralization, including *ZIC3*, *LEFTY1*, *ACVR2B*, *CFC1*, and *NKX2.5*, have been shown to be associated with these syndromes. Asplenia is also part of the Stormorken syndrome, an autosomal dominant disorder that is due to heterozygous gain-of-function mutations of *STIM1*, and which is further characterized by thrombocytopenia with platelet dysfunction, myopathy, miosis (constriction of the pupils), and ichthyosis (scaly skin). Autosomal recessive loss of function mutations of the *STIM1* gene are involved in a primary immune deficiency with defective Ca^{2+} flux and immune dysregulation (see Case 55).

The second category is isolated congenital asplenia. This is a group of conditions in which the only abnormality is the lack of splenic tissue. Most of the patients with isolated congenital asplenia were diagnosed after episodes of overwhelming pneumococcal infections, either post mortem or while screening family members of affected individuals. It has been estimated that the incidence of congenital asplenia is 0.5–1 case per million births. The majority of patients with isolated congenital asplenia carry heterozygous mutations in the ribosomal protein SA (*RPSA*) gene, causing haploinsufficiency. In affected families, the disease is fully penetrant.

After the events described in this case, the Vanderveers had three more children. One of the boys and the girl were also born without a spleen; the other boy had a normal spleen. This family provides us with an uncomplicated circumstance in which to examine the role of the spleen in the immune response against bacteria. In particular, the marginal zone (MZ) of the spleen (see Fig. 30.2) contains

Fig. 30.5 A scintillation scan of the abdomen after intravenous injection with radioactive colloidal gold (^{198}Au) reveals that Betsy and David Vanderveer have no spleens. The top panel shows an abdominal scan of Betsy's mother. The large mass on the left is the liver and the small mass on the right is the spleen. The reticuloendothelial cells of both liver and spleen take up the labeled gold within 15 minutes after the injection. No spleen is seen in either Betsy (middle panel) or David (bottom panel).

specialized B lymphocytes (MZ B cells) and macrophages that play a critical role in protecting against blood-borne pathogens. MZ B cells are also important for the production of antibodies against polysaccharide antigens, such as those that are present on the surface of encapsulated bacteria. Consequently, the absence of the spleen is associated with increased susceptibility to bacteremia, usually caused by the encapsulated bacteria *Streptococcus pneumoniae* or *Haemophilus influenzae*. This susceptibility is caused by a failure of the immune response to these common extracellular bacteria when they enter the bloodstream.

Surgical removal of the spleen is quite common. The capsule of the spleen may rupture from trauma, for example in an automobile accident. In such cases, the spleen has to be removed very quickly because of blood loss into the abdominal cavity. The spleen may also be removed surgically for therapeutic reasons in certain autoimmune diseases, or because of a malignancy in the spleen. After splenectomy, patients, particularly children, are susceptible to bloodstream infections by microorganisms against which they have no antibodies. Microorganisms against which the host has antibodies are removed quickly from the bloodstream by the liver, where the Kupffer cells complement the role of the red pulp of the spleen.

Finally, some conditions are characterized by functional asplenia. In these situations, the spleen is present but its function is impaired. Examples include sickle cell anemia, thalassemia, and spleen infarcts.

Patients with anatomical or functional asplenia are highly prone to overwhelming infections sustained by encapsulated bacteria. To reduce the risk of these infections, they need to be actively immunized against *S. pneumonia*, *H. influenzae* type b, and *Neisseria meningitidis*, possibly using protein-conjugated vaccines. Moreover, specific precautions, including prophylactic antibiotics, are recommended for most individuals with an absent or non-functional spleen.

Questions.

- 1 Nicholas Biddleboy, a 5-year-old boy, has had his spleen removed after a sledding accident, during which both he and his sled struck a tree trunk. In the emergency room of a nearby hospital, it was determined that his spleen had ruptured. The surgeon, after removal of a spleen that had indeed ruptured, calls you for an immunology consultation. What do you advise?
- 2 The Vanderveer family is unique in the medical literature. The parents, who were distantly related, were normal and had normal spleens. Five of their eight children were born without spleens. Of these, only Betsy subsequently had children—four boys and one girl. They are all normal and have spleens. What is the inheritance pattern of congenital asplenia in this family? According to Mendelian laws how many of the eight Vanderveer children would be expected to have no spleen?